



## ORIGINAL RESEARCH

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# Emergency Department-Based Medication for Opioid Use Disorder: A 5-Year Experience

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### Abstract

**Introduction:** Medication for opioid use disorder (MOUD) programs are effective in treating opioid use disorder (OUD). Successful programs are resource-intensive. We describe our 5-year emergency department (ED)-based MOUD program results.

**Methods:** Our MOUD program uses Addiction Care Coordinators to coordinate care for OUD patients presenting to our ED. Peer recovery specialists provide counseling. Our MOUD program is available daily at our 2 hospitals. In 2021, we expanded our MOUD program to our 2 free-standing EDs using a telemedicine platform. Outcomes included the number of patients inducted over 5 years (2019-2023), the number inducted through telehealth, and the retention rate in the MOUD outpatient program at 1 and 6 months.

**Results:** All 373,797 ED patients were screened for OUD over the 5 years, and 1072 (2.8%) were inducted into the MOUD program. A total of 53 patients (4.9%) were inducted through telemedicine. The number of patients inducted declined yearly; however, inpatient admissions for OUD declined less. The overall retention rate was 36.5% at 1 month and 28.8% at 6 months and then declined yearly. According to state data, the number of opioid and buprenorphine prescriptions also went down during this 5-year period.

**Discussion:** ED-based MOUD programs expand access to treatment for patients with OUD. Patients' use of the ED-based MOUD program has declined each year, and retention rates have fallen. Several factors could be involved, such as increased strength of fentanyl, worsening withdrawal symptoms, increased inpatient admissions, and expansion of MOUD to community sites. Cost-effective strategies are needed to continue these programs.

**Keywords:** opioid use disorder, medication for opioid use disorder, MOUD, telehealth

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## 1 INTRODUCTION

### 1.1 Background

The United States has seen a significant increase in opioid drug use, leading to the opioid crisis. In 2020, an estimated 2.7 million people aged 12 years or older in the United States had an opioid use disorder (OUD).<sup>1</sup> The total annual OUD-related costs to the United States in 2018 were \$786.8 billion to society, \$93 billion to taxpayers, and \$89.1 billion to the health care sector.<sup>2</sup> The mean value of preventing a patient from having an OUD across all ages was \$2.2 million, \$325,125, and \$244,030 from the societal, taxpayer, and health care sector perspectives, respectively.<sup>2</sup> Despite the attention OUD has received, opioid-involved overdose deaths rose from 21,088 in 2010 to 107,941 in 2022.<sup>3</sup>

### 1.2 Importance

For patients with OUD, the emergency department (ED) represents a critical access point for receiving medical care and, thus, a significant opportunity to help OUD patients. Medication for opioid use disorder (MOUD) has become the state-of-the-art treatment modality for ED management of OUD. MOUD decreases mortality, reduces overdoses, increases treatment retention, and decreases the costs associated with addressing the opioid crisis epidemic.<sup>4–8</sup> Initial randomized control trials to evaluate the use of an ED-based MOUD found that the ED can be utilized to provide initial buprenorphine inductions, resulting in modest short-term treatment retention rates.<sup>9–13</sup>

### 1.3 Goals of This Investigation

This study reports on our 5-year experience with an ED-based MOUD program, specifically focusing on telehealth expansion, treatment trends, and patient retention rates.

## 2 METHODS

### 2.1 Program Description

Summa Health's *First Step* program is an innovative ED-based MOUD that began in October 2018 at the 2 Summa Health System's EDs.<sup>10</sup> In 2020, the program began utilizing the Screening, Brief Intervention, Referral to Treatment (SBIRT) evidence-based model to treat patients.<sup>14</sup> The transitional care lynchpins in the *First Step* Program are the Addiction Care Coordinators (ACCs), who are registered nurses who guide the patients through their recovery journeys. The ACCs provide traditional nursing services such as physical assessments, withdrawal screening, medication administration, and patient education; however, their singular focus on the substance use disorder (SUD) patient population makes them distinctive. Care coordination is arguably the most critical function of the position, and this navigational responsibility ensures that patients stay on

course while navigating a complex (and sometimes discouraging) health care environment. ACCs provide assessments based on a framework developed by the American Society of Addiction Medicine (ASAM)<sup>15</sup> and create individualized treatment plans with patients, which include follow-up plans that consider numerous social determinants of health (ie, ASAM level of care, geographic location, type of health insurance, access to care, and patient preference).

Furthermore, the ACCs evaluate any relevant comorbid conditions and connect patients to medical follow-up if necessary. The ACCs also facilitate a warm handoff of impending referrals to internal and external service entities. Along with the necessary clinical referrals, the ACCs are also essential in connecting patients to recovery support services, such as peer recovery specialists (PRS) from Catholic Charities, who provide on-call services (24/7) to the EDs. Most follow-up appointments (96%) occur within the Summa Health System. Four percent are referred to external community MOUD programs. Harm reduction is also part of our program's treatment approach, including ACC referrals to needle exchange programs and the distribution of Naloxone kits. In conjunction with the PRS, the ACCs provide outreach calls to *First Step* patients to discuss their progress, re-engage if the patient has not followed through with treatment, assist with psychosocial barriers to treatment, and encourage those on their path to recovery.

If the patient agrees to MOUD, treatment with buprenorphine is initiated while in the ED. Treatment consists of 8 mg/2 mg buprenorphine/naloxone for one dose, followed by reevaluation 1 hour later, and repeated dosing if the Clinical Opiate Withdrawal Score exceeds 8. If urgent (next day) addiction medicine follow-up is unavailable, patients are scheduled to return to the ED the next day for bridge treatment. This is primarily to help coordinate care, maintain the patient in the MOUD program, and reduce the diversion of buprenorphine. For patients arriving outside ACC hours (11 PM to 11 AM), buprenorphine is given at the physician's discretion, with an automatic referral sent to our ACCs for follow-up coordination the next day. The process for treatment and referrals is the same for hospital and free-standing ED (FSED) OUD patients. The maximum distance between the hospital referral clinics and either FSED is 14 miles. Summa Health's expansion efforts developed a hybrid strategy focused on program sustainability. In 2021, our program used telehealth technology to provide addiction care coordination, including case management services, to our 2 FSEDs in place of its in-person practices at the 2 hospital sites, reducing personnel expenses significantly.

### 2.2 Measurements

Our MOUD program received multiple grants to begin and sustain the program, including the United Way, the Ohio Department of Mental Health and Addiction Services, the Federal Communication Commission, and the Centers for Disease Control and Prevention (CDC). Metrics measured

## The Bottom Line

Emergency department-based medication for opioid use disorder can be expanded through telemedicine. However, utilization and retention rates declined over the 5-year study period. Cost-effective strategies are needed to continue these resource-intensive programs.

included the total number of patients screened, number of patients referred into the MOUD program, number inducted in the ED, number who received naloxone kits, and retention rates into a behavioral health program at 1 and 6 months. We also separately measured the enrollment of patients into the MOUD program at our FSEDs through telehealth. Since this was part of a quality improvement project, the Summa Health System Institutional Review Board found it to be exempt from review.

### 2.3 Analysis and Interpretation

Data were collected on the number of ACCs trained and the number of patients enrolled in standard face-to-face MOUD and the number enrolled by the telehealth MOUD. We also compared the number of patients retained in treatment at 1 and 6 months and examined the follow-up visit rate between the 2 platforms. All measures were tabulated as counts and percentages for categorical variables. Means, SDs, medians, and IQRs were analyzed for continuous variables. Mean differences among subgroups were measured using chi-square tests for categorical variables and analysis of variance along with distribution-free nonparametric tests with significance set at  $P < .05$ . Data were analyzed using IBM SPSS Statistics (Version 29, International Business Machines Corporation).

### 3 RESULTS

A total of 373,797 ED patients were screened for OUD over the 5 years, and 1072 (2.8%) were inducted into the MOUD program during their ED visit. We had a total of 53 patients (4.9%) inducted through our telemedicine MOUD program. All patients had OUD, and 22.8% had polysubstance abuse (62.4% methamphetamine, 15.3% benzodiazepine, 11.8% cocaine, 8.4% alcohol, 2.1% other). The mean age was 32.3 years (range, 20-62 years), and 65% identified as male, whereas 35% identified as female. Of the patients inducted, 93% were White, 6% were Black, and 1% were other. Insurance status was 68% had Medicaid, 19% Medicare, 9% private insurance, 14% self-pay, and 1% other (Table). In our study population, 27% were employed, and over 73% had at least a high school education.

The number of patients inducted declined yearly; however, hospital admissions for OUD declined less (Fig 1). The overall retention rate was 36.5% (95% CI: 36.1%-36.9%) at 1 month and 28.8% (95% CI: 28.4%-29.2%) at 6 months but declined yearly (Fig 2). There was no difference in retention rates based on induction (hospital, telehealth, or follow-up site). The retention rate for the telemedicine MOUD cohort at the 2 FSED was 34.3% (95% CI: 32.5%-36.1%) at 1 month and 26.6% (95% CI: 24.9%-28.3%) at 6 months and was similar to the in-person MOUD at the 2 hospitals ( $P = .087$  and  $.069$ , respectively). Retention rates at 1 and 6 months significantly declined from the program's first 2 years (47.7% and 29.1%, respectively) to the last 3 years (35.5% and 24.4%,  $P < .001$ ) (Fig 2). According to state data,<sup>16</sup> the number of opioid (Fig 3) and buprenorphine (Fig 4) prescriptions in the 5-county referral area also declined over the 5-year period. Opioid overdose deaths also remain high in Ohio but slightly declined in 2023 (Fig 5). A total of 1298 Naloxone kits were given out in the ED over the 5 years.

### 4 LIMITATIONS

The current study has several limitations. First, this study was done at a single health system in Ohio. Although our MOUD program follows Substance Abuse and Mental Health Services (SAMHSA) best practices, it is extensive and resource-intensive. This may limit the external validity of our results; however, health systems without such a robust program may even have less retention in their MOUD programs. Second, we attempted to follow-up all patients at 1 and 6 months. Some patients were lost to follow-up. We tried to account for this by including patients in an intention-to-treat assignment. Hence, patients whom we were unable to contact or follow up with were counted as not being retained in the MOUD program. Lastly, this study did not assess patients' readiness or internal motivation to quit at baseline over the entire 5-year period. In 2022-2023, we conducted psychological surveys at induction to determine whether certain tests could predict retention. This study, which found that patients with less posttraumatic stress and higher income were more likely to remain in treatment, is currently being reviewed for publication.

### 5 DISCUSSION

In 2023, the CDC reported that Ohio ranked high in the nation in the number of unintentional overdoses at 4739.<sup>17</sup> Summit County, where Summa Health is located, ranked fifth of the 88 counties in Ohio with 230 overdose deaths in 2022.<sup>18</sup> Patients seeking treatment were often met with delays in access up to 10 days.<sup>19</sup> Not only was the wait time a barrier, but the fragmented network of addiction treatment facilities proved difficult to navigate. Frustrated patients often found themselves continuing to use to avoid withdrawal. The risk of unintentional overdose continued.

**TABLE.** Descriptive statistics.

| Characteristics     | Hospital MOUD<br>(N = 1019) (95% CI) | Tele-MOUD<br>(N = 53) (95% CI) | P     |
|---------------------|--------------------------------------|--------------------------------|-------|
| Age, y              | 32.3 (31.8, 32.8)                    | 31.8 (29.4,34.1)               | .686  |
| Age group, y        |                                      |                                |       |
| 18-29               | 83.6% (83.0, 84.2)                   | 82.6% (79.5, 85.7)             | .566  |
| 30-59               | 15.6% (15.3, 15.9)                   | 15.5% (14.4, 16.6))            | .882  |
| >60                 | 0.79% (0.75, 0.83))                  | 1.9% (1.79, 2.0)               | .0001 |
| Sex at birth        |                                      |                                |       |
| Female              | 34.1% (33.7, 34.5)                   | 35.8% (33.7, 37.8)             | .121  |
| Male                | 65.9% (65.4, 66.4)                   | 64.2% (62.1, 66.3)             | .141  |
| Race/ethnicity      |                                      |                                |       |
| White, non-Hispanic | 92.3% (91.8, 92.8)                   | 90.6% (88.3, 92.9)             | .174  |
| Black, non-Hispanic | 5.9% (5.8, 5.9)                      | 7.5% (6.9, 8.1)                | .0001 |
| Other               | 1.8% (1.7, 1.8)                      | 1.9% (1.8, 2.0)                | .368  |
| Primary payer       |                                      |                                |       |
| Medicaid            | 57.2% (56.7, 57.6)                   | 58.5% (57.0, 60.6)             | .199  |
| Medicare            | 18% (17.6, 18.4)                     | 19% (17.3, 20.7)               | .267  |
| Uninsured           | 14% (13.6, 14.4)                     | 15.1% (13.5, 16.7)             | .186  |
| Other               | 10.8% (10.7, 10.9)                   | 7.4% (6.9, 7.9)                | .0001 |
| Retention rate      |                                      |                                |       |
| 1 mo                | 36.5% (36.1, 36.9)                   | 34.3% (32.5, 36.1)             | .087  |
| 6 mo                | 28.8% (28.4, 29.2)                   | 26.6% (24.9, 28.3)             | .069  |

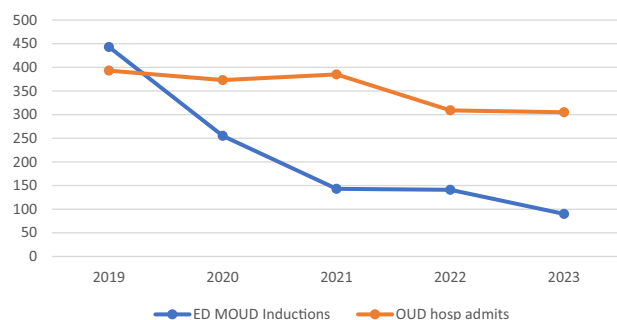
MOUD, medication for opioid use disorder.

We launched our ED-based MOUD program through grant and philanthropic funding in February 2018. Our program, based at 2 system hospitals, afforded the surrounding communities access to an entire spectrum program that consisted of evidence-based treatment options and recovery support services. A diagnosis of OUD compromises every aspect of a patient's life; thus, a multifaceted treatment approach is required to address the critical biopsychosocial needs of the individual. Patients with OUD and SUD have higher rates of ED visits and hospital readmissions.<sup>20</sup> When left untreated,

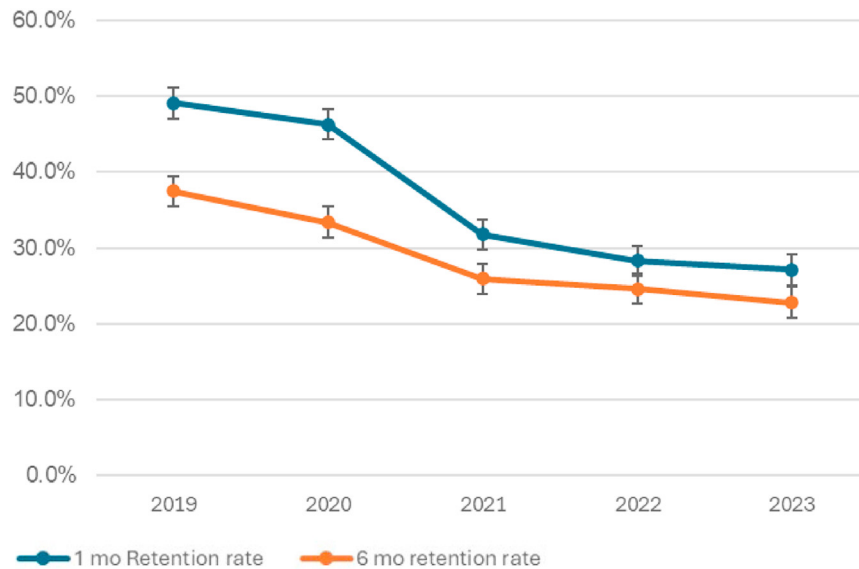
SUDs are associated with high rates of injury, disability, and death.<sup>21</sup> Therefore, a robust care coordination network is crucial in navigating a fragmented health care landscape and meeting patients' addiction, behavioral health, and medical requirements and to provide them with the best opportunity on their road to recovery.

Our MOUD program, administered within the system's EDs located in one urban teaching hospital ED and one community hospital ED, was effectively expanded to 2 FSEDs. The provision of program services, including addiction care coordination, was realized via the implementation of telehealth technology. On-call peer recovery support was extended to the FSEDs. The MOUD project team also mobilized a strategic education campaign consisting of an updated addiction curriculum and in-person training seminars to facilitate staff understanding and competence. The compendium of education delivered to personnel at the FSEDs focused on SBIRT, outcomes tracking, data collection, cultural competencies, and proper equipment use. Patient care coordination, tracking, and assessment were expanded and assessed to ensure alignment across all sites.

Innovative ideas, lean processes, cost-reduction initiatives, and value-based contracts are needed to allow our MOUD program to reach more patients without increasing expenses. The current model has been shown to have improved long-term retention but is labor-intensive, resulting in



**FIGURE 1.** OUD patients admitted versus ED-based MOUD. ED, emergency department; MOUD, medication for opioid use disorder; OUD, opioid use disorder.

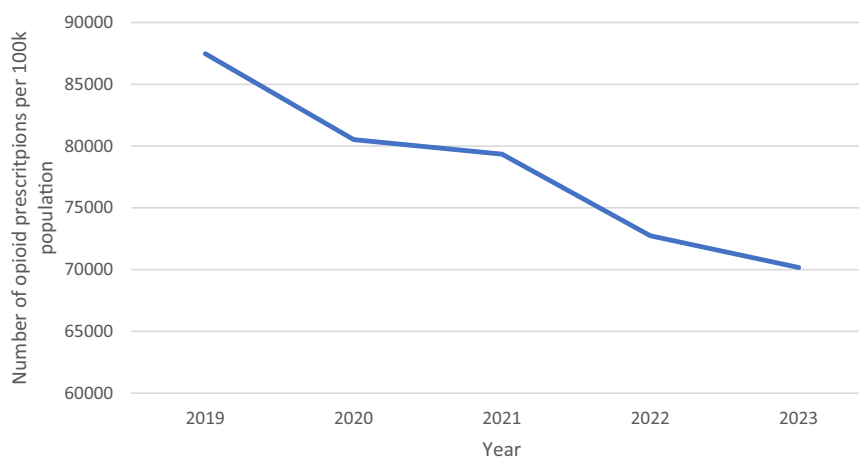


**FIGURE 2.** ED-based MOUD retention rates with 95% CIs. ED, emergency department; MOUD, medication for opioid use disorder.

unsustainable costs. Finding an enduring and financially viable approach for treating OUD patients was critical. Through the addition of telehealth technology, our MOUD program was able to offer ACC services to our 2 FSEDs, thereby improving the program's financial sustainability.

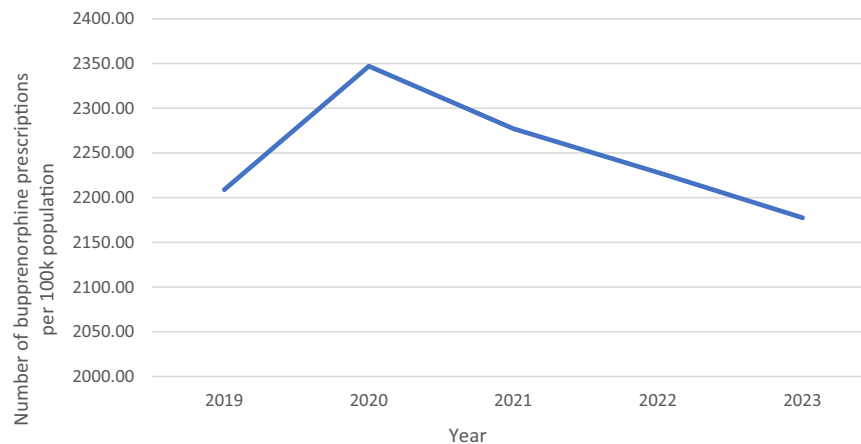
Our results noted success in screening patients for OUD and inducting them into our ED-based MOUD program. The initial 2 years of the program had 698 patients inducted and retention rates of 47.7% and 29.1% at 1 and 6 months, respectively. However, the last 3 years had a decrease in MOUD inductions despite the increase in SBIRT screening protocols, with 374 inductions and retention rates of 35.5% and 24.4% at 1 and 6 months, respectively. The reasons for this are unclear, but several factors could be considered. First, the primary opioid used in our area was fentanyl. Some have

postulated that MOUD may be less effective due to the increased potency of fentanyl.<sup>22</sup> Additionally, it is believed that fentanyl use may have worsening withdrawal symptoms with buprenorphine, making MOUD treatment less palatable and leading to increased inpatient admission for OUD or other alternatives such as methadone.<sup>23</sup> The expansion of MOUD options at other community sites, including primary care offices, may have reduced the number of patients in our specific ED-based MOUD program.<sup>24,25</sup> The COVID pandemic occurred during the study period, but the ED census returned to 2019 levels by 2022, so we do not believe this affected MOUD enrollment. Lastly, the marked increase in hospital crowding and subsequent inpatient boarding may have affected MOUD enrollment and retention as patients may not have wanted to wait for their MOUD treatment.



**FIGURE 3.** Opioid prescriptions written per 100,000 population in 5 referral county areas.



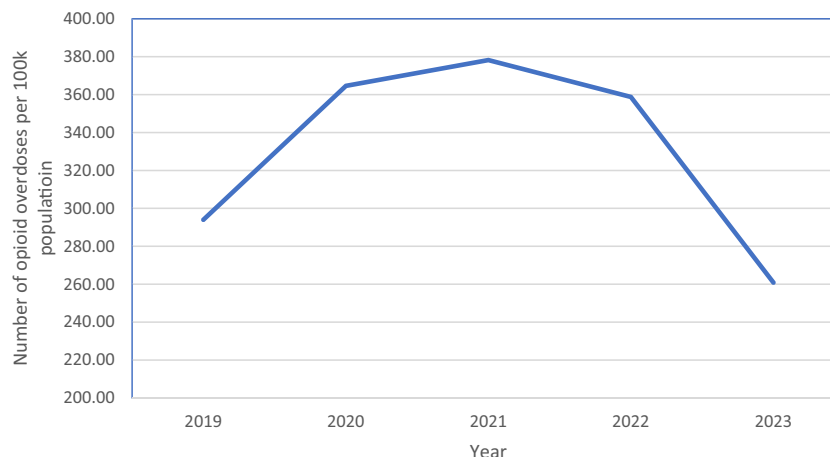


**FIGURE 4.** Buprenorphine prescriptions written per 100,000 population in 5 county referral areas.

The cost benefit of such expansive ED-based MOUD programs may be called into question, especially in light of new studies that show evidence-based practices for reducing opioid-related overdose deaths, including overdose education and naloxone distribution, the use of medications for the treatment of OUD, and prescription opioid safety are not decreasing opioid overdose deaths.<sup>26</sup> Efforts to make these programs more clinically and cost-effective need to be developed, such as regionalization of such programs and increased clinician acceptance. The problem facing EDs is that MOUD is an effective therapy, but its utilization by patients is declining. The causes of this may be multifactorial, so innovative solutions are needed to keep this valuable treatment option available. Program costs are also a concern due to the intensive nature of such programs, particularly the direct cost of ACCs. Although our program was initially funded by Department of Health grants, these have since ended. We have expanded our program to the inpatient hospital setting and primary care clinic in our health care system to continue our program. Cost-effectiveness can be improved by reducing

expenses by keeping OUD patients in therapy and increasing behavioral health revenue to offset program costs. Additional ideas for enhancing the cost viability of these programs include developing a hub-and-spoke model for addressing the complex care of OUD.<sup>27</sup> This model would create a regional specialty treatment center, using the ACCs at our facility, which has the resources to treat OUD and the complex psychosocial issues. The spokes would be outpatient addiction centers, primary care offices, and community EDs that could use the ACCs and our specialized resources to help treat their patients. We have proposed this to the State Department of Health, but there are still challenges with collecting revenue for these services. The best justification for this model lies in reducing health care costs, ED visits, and morbidity and mortality in OUD patients under this model.

In conclusion, ED-based MOUD programs can be effective in treating OUD by expanding access to valuable treatment on the front line of health care. MOUD treatment can also be expanded to other sites, such as FSEDs, through telehealth. Over our 5-year study period, inductions in our



**FIGURE 5.** Opioid overdose deaths per 100,000 population in 5 county referral areas.

MOUD fell during the last 3 years, and retention rates in a long-term behavioral health program also declined. New strategies will need to be developed to continue to support our MOUD program and maintain treatment engagement for patients with OUD.

## AUTHOR CONTRIBUTIONS

DCS, JM, LH, SV, and NJ conceived the study, designed the trial, and obtained research funding. DCS, JM, LH, NJ, and JK supervised the conduct of the trial and data collection. LH, JM, JK, and SV undertook recruitment of participating centers and patients and managed the data, including quality control. DCS, JO, LH, and JCM provided statistical advice on study design and analyzed the data; JM chaired the data oversight committee. DCS drafted the manuscript, and all authors contributed substantially to its revision. DCS takes responsibility for the paper as a whole.

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## CONFLICT OF INTEREST

All authors have affirmed they have no conflicts of interest to declare.

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